

General User Guide

This guide explains how to use VirtualController to play any supported game with one or more players.

What You Need

- A Windows PC running the Game Library Launcher
- GameServer running on the same PC or on the network
- One or more input devices, either of which:
 - Keyboard
 - Physical controller
 - Mobile phone controller

Start the Server

1. Open a terminal where you want the server to run.
2. Run the server:

```
.\build\bin\Release\GameServer.exe
```

If the server is on another PC, note its IP address and port.

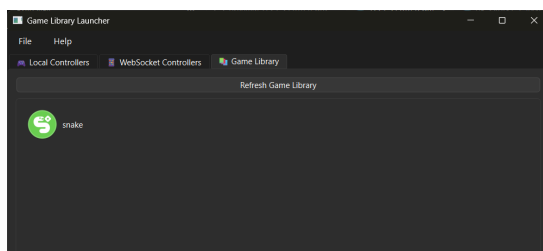
Launch the Game Library

1. Start the launcher:

```
.\build\bin\Release\GameLibraryLauncher.exe [ip/hostname] [port]
```

The default server port is 8765.

2. The Game Library tab lists available games.

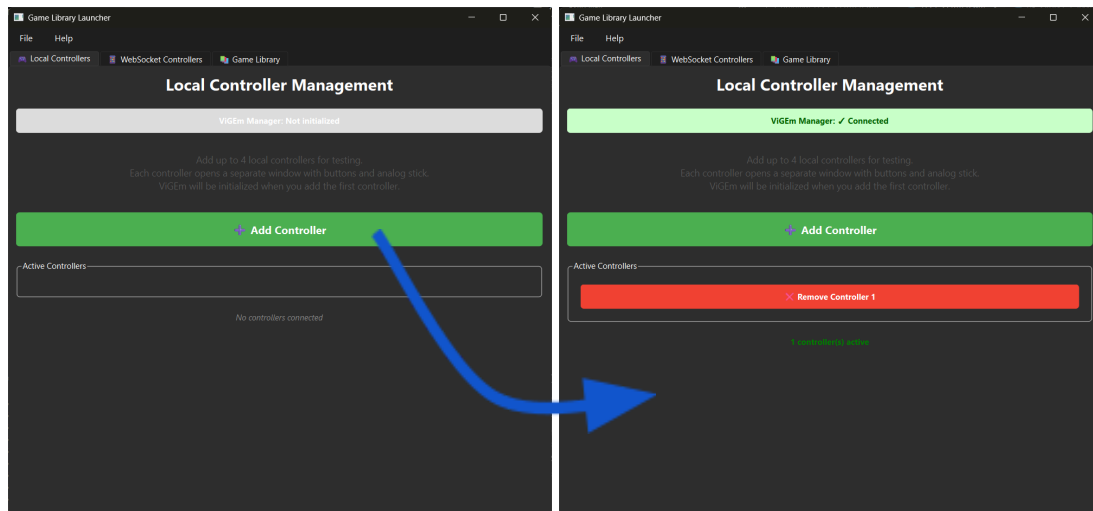


Game launcher displaying library

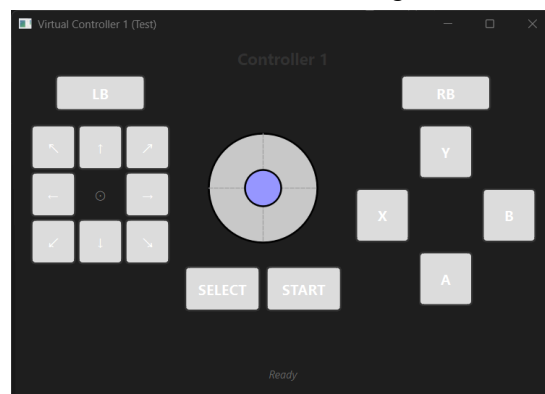
Add Local Controllers

Use this when you want to create virtual controllers on the PC (ViGEm).

1. Open the **Local Controllers** tab.
2. Click **Add Controller**.
3. A test controller window appears and a new controller becomes active.



Game launcher before / after adding a local controller

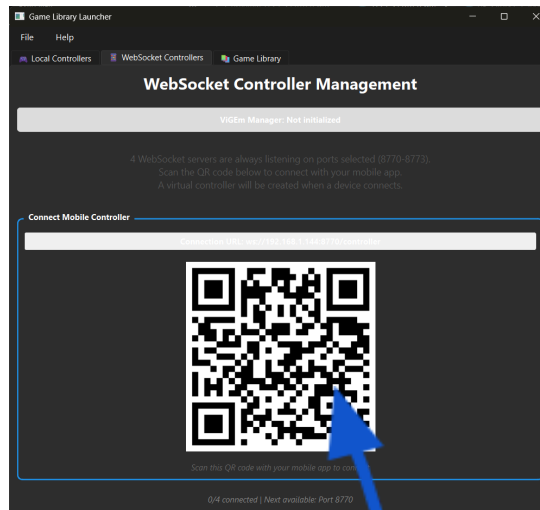


Local controller opening in a new window

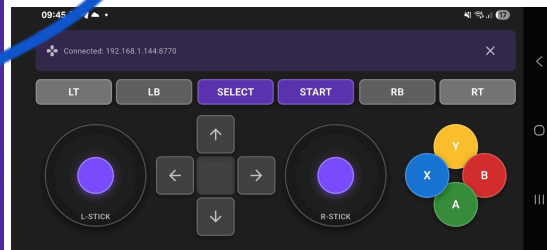
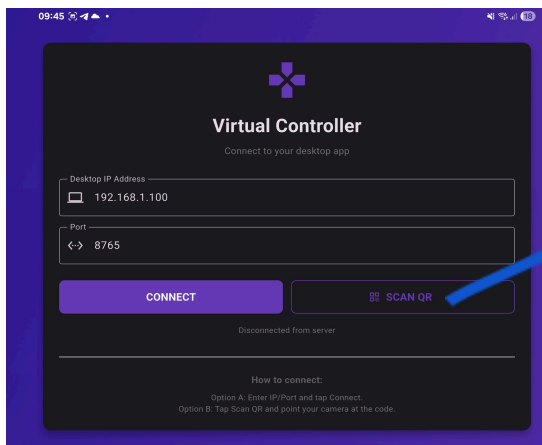
Add Mobile Controllers

Use this when you want phones to act as controllers.

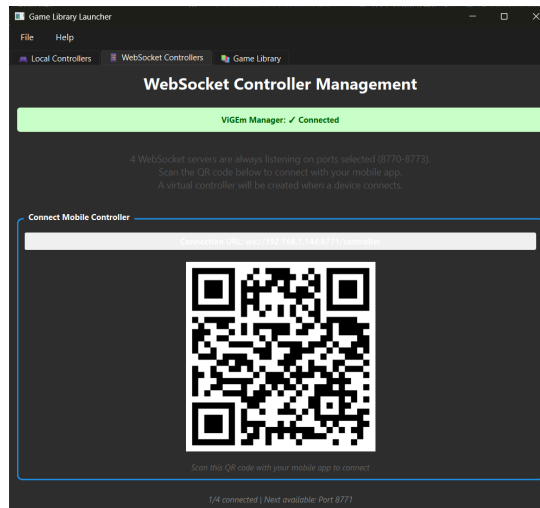
1. Open the **WebSocket Controllers** tab.
2. If ViGEm is connected, a QR code is displayed.
3. On your phone, open the mobile controller app.
4. Enter the PC IP and port (or scan the QR code) to connect.



Game launcher before adding a mobile controller



Virtual controller phone app connecting to the computer



Game launcher after adding a mobile controller

Launch a Game

1. In the **Game Library** tab, click a game (for example, snake).
2. The game starts in its own window and connects to the server.

If the game shows a lobby screen, wait for all players to connect, then start the game from inside the game window.

Play with Multiple Players

- Each connected device counts as one player.
- You can mix keyboard, controllers, and mobile devices.
- The server assigns a unique player ID to each connected device.

Troubleshooting

- **Game does not connect:** Verify GameServer is running and the IP/port passed to the launcher or game is correct.
- **Mobile controller cannot connect:** Make sure the phone and PC are on the same network and the QR code URL uses the PC IP address.
- **No games listed:** Ensure the game executables exist in `build/bin/Release/` and restart the launcher.

Quick Checklist

- Server running
- Launcher running
- At least one input device connected
- Game launched from Game Library